

CST8390 - Lab 1 Answers

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Student Number: 041219844

During demo time, you should be able to answer questions on the process that you did in this lab.

When you take screenshots, make sure that you are getting the shot of the full Weka.

After filling the answers, save this file as a pdf and upload it to the Brightspace.

Part 1 - Data Exploration in Weka

Flower Type	Count	Attribute	Min	Max	Mean	StdDev
Iris Setosa	50	sepalength	4.3	7.9	5.843	0.828
Iris versicolor	50	sepalwidth	2	4.4	3.054	0.4341
Iris virginics	50	petallength	1	6.9	3.759	1.764
		petalwidth	0.1	2.5	1.199	0.763

Part 2 - Data Preparation and Cleaning

3. Student number: 041219844

Row number of the new instance: 45

Salary: 19844

Screenshot:

42	50010150	Burton	Dudden	bdudden39@japanpost.jp	050 Nova Court	Mexico	4 INR	44444
43	50010151	Rodge	Sarfass	rsarfass3d@wired.com	82 7th Hill	Mexico	4 MXD	15999
44	50010152	Alison	Lampert	alamperte@list-manage.com	78 Express Plaza	U.S.A.	1 USD	27000
45	41219844	Hiralben	Prajapati	Hiralben_Prajapati@test.com	412 Hiralben Street	Canada	10 CAD	19844
46	50010153	Yulma	Peyntue	ypeyntue26@mayoclinic.com	3257 American Crossing	China	3 CHY	150000
47	50010154	Reade	McCumes	rmccumesky2y@list-manage.com	6766 Schmedeman Road	China	3 CHY	150000
48	50010155	Velma	Andraud	vandraud1x@issuu.com	58 Harper Pass	Germany	2 EUR	96300

Answer: Country, Branch, Currency, Salary

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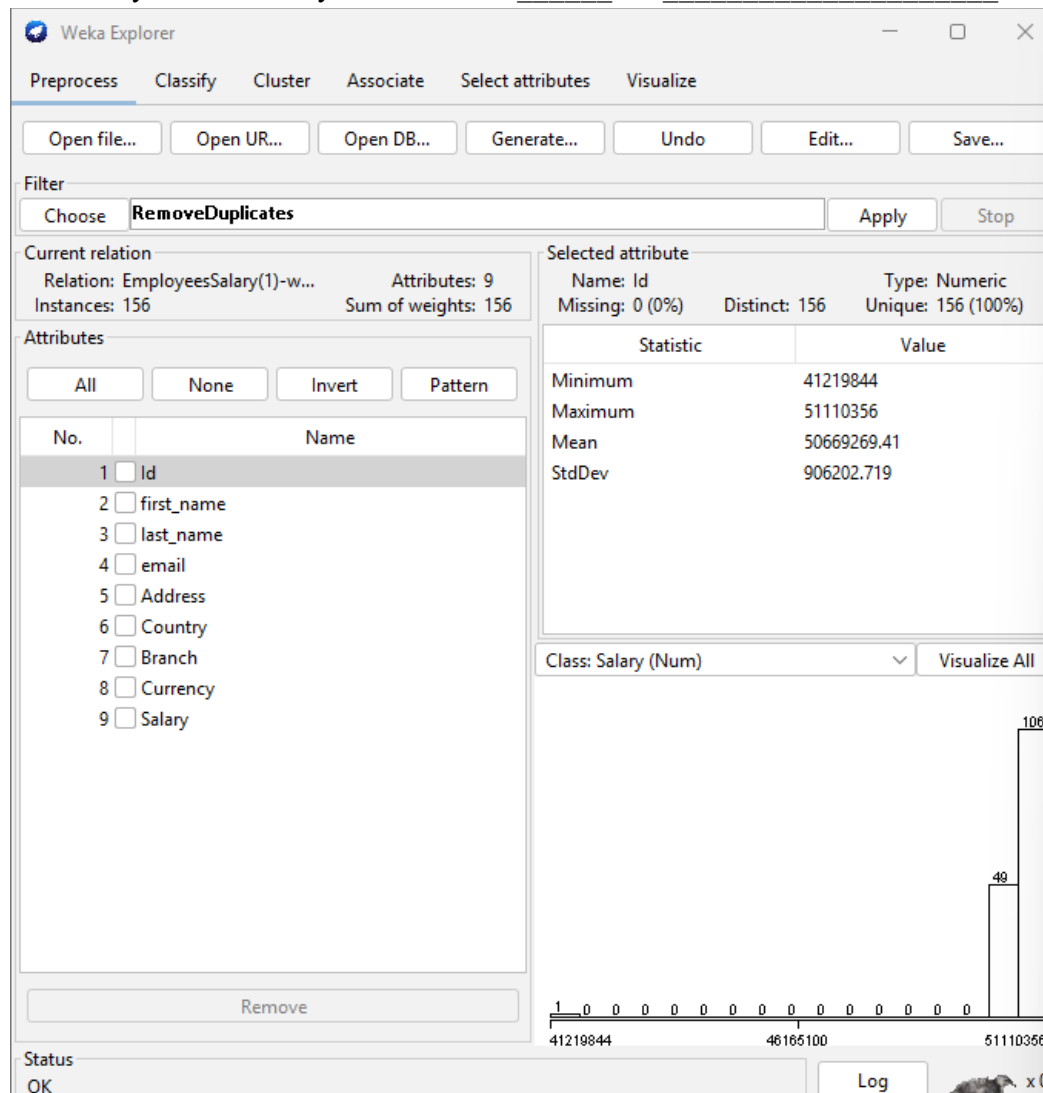
Answer: Country, Branch, Currency, Salary

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Answer: Country, Branch, Currency, Salary

(You must find at least 10 anomalies to get full marks)

10. How many instances do you have now? _____ 156 _____



15.

a. How many instances do you have now? _____ 151 _____

b. How many duplicates (how many got removed): _____ 5 _____

17. Take a screenshot and paste it here.

Weka Explorer

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□

×

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Open file...

Open UR...

Open DB...

Generate...

Undo

Edit...

Save...

Filter

Choose

RemoveDuplicates

Apply

Stop

Current relation

Relation: EmployeesSalary(1)-w...
Instances: 151

Attributes: 8
Sum of weights: 151

Selected attribute

Name: first_name
Missing: 0 (0%)

Type: String
Distinct: 149
Unique: 147 (97%)

Attributes

All

None

Invert

Pattern

No.		Name
1	☒	first_name
2	☐	last_name
3	☐	email
4	☐	Address
5	☐	Country
6	☐	Branch
7	☐	Currency
8	☐	Salary

Remove

Class: Salary (Num)

Visualize All

Attribute is neither numeric nor nominal.

Status

OK

Log

 x 0

Nominal to Binary

18. How many nominal attributes do you have? 3 (Country, Branch, and Currency)
20. Take a screenshot and paste it here.

The screenshot shows the Weka Explorer interface with the 'Preprocess' tab selected. The 'Filter' dropdown is set to 'NominalToBinary -R first-last'. The 'Current relation' is 'EmployeesSalary(1)-w...'. The 'Attributes' list shows 23 attributes, with 'Country=China' selected. The 'Selected attribute' panel shows statistics for 'Country=China': Minimum 0, Maximum 1, Mean 0.238, and StdDev 0.428. The 'Class' is set to 'Salary (Num)'. A bar chart at the bottom shows the distribution of the 'Country=China' attribute, with a bar at 0 having a count of 115 and a bar at 1 having a count of 36.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open UR... Open DB... Generate... Undo Edit... Save...

Filter

Choose **NominalToBinary -R first-last** Apply Stop

Current relation

Relation: EmployeesSalary(1)-w... Attributes: 23
Instances: 151 Sum of weights: 151

Attributes

All None Invert Pattern

No.	Name
1	☐ first_name
2	☐ last_name
3	☐ email
4	☐ Address
5	☒ Country=China
6	☐ Country=U.S.A.
7	☐ Country=Germany
8	☐ Country=Mexico
9	☐ Country=Canada
10	☐ Country=Japan
11	☐ Branch=1
12	☐ Branch=2
13	☐ Branch=3
14	☐ Branch=4
15	☐ Branch=6
16	☐ Branch=10
17	☐ Currency=CHY

Remove

Selected attribute

Name: Country=China Type: Numeric
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.238
StdDev	0.428

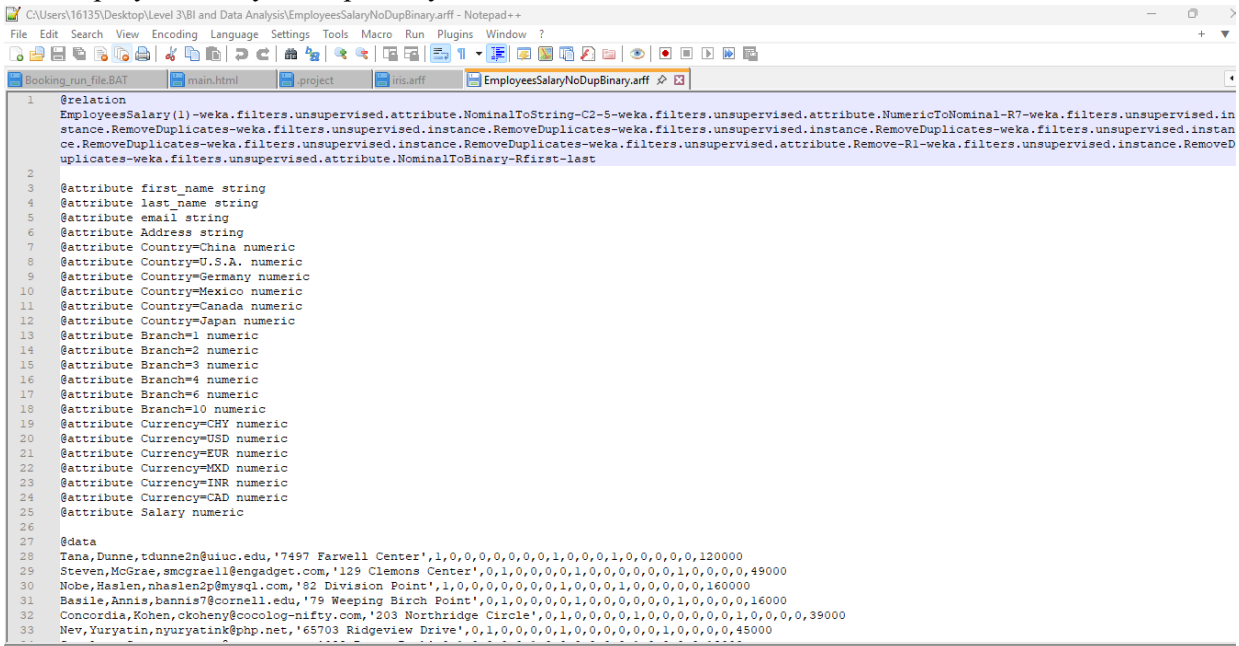
Class: Salary (Num) Visualize All

115 36

0 0.5 1

Status OK Log x 0

23. Take a screenshot of the file while it is opened in Notepad++. Header should be visible.
EmployeesSalaryNoDupBinary.arff::



```
1 @relation
2 EmployeesSalary(1)-weka.filters.unsupervised.attribute.NominalToString-C2-5-weka.filters.unsupervised.attribute.NumericToNominal-R7-weka.filters.unsupervised.in
3 stance.RemoveDuplicates-weka.filters.unsupervised.instance.RemoveDuplicates-weka.filters.unsupervised.instance.RemoveDuplicates-weka.filters.unsupervised.instan
4 ce.RemoveDuplicates-weka.filters.unsupervised.instance.RemoveDuplicates-weka.filters.unsupervised.attribute.Remove-R1-weka.filters.unsupervised.instance.RemoveD
5 uplicates-weka.filters.unsupervised.attribute.NominalToBinary-Rfirst-last
6
7 @attribute first_name string
8 @attribute last_name string
9 @attribute email string
10 @attribute Address string
11 @attribute Country=China numeric
12 @attribute Country=U.S.A. numeric
13 @attribute Country=Germany numeric
14 @attribute Country=Mexico numeric
15 @attribute Country=Canada numeric
16 @attribute Country=Japan numeric
17 @attribute Branch=1 numeric
18 @attribute Branch=2 numeric
19 @attribute Branch=3 numeric
20 @attribute Branch=4 numeric
21 @attribute Branch=6 numeric
22 @attribute Branch=10 numeric
23 @attribute Currency=CHY numeric
24 @attribute Currency=USD numeric
25 @attribute Currency=EUR numeric
26 @attribute Currency=MXD numeric
27 @attribute Currency=INR numeric
28 @attribute Currency=CAD numeric
29 @attribute Salary numeric
30
31 @data
32 Tana,Dunne,tdunne2n@uiuc.edu,'7497 Farwell Center',1,0,0,0,0,0,0,1,0,0,0,1,0,0,0,0,120000
33 Steven,McOree,smcraell@engadget.com,'128 Clemens Center',0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,1,0,0,0,0,49000
34 Nobe,Haslen,nhaslen2p@mysql.com,'82 Division Point',1,0,0,0,0,0,0,1,0,0,0,1,0,0,0,0,0,0,0,0,160000
35 Basile,Annis,bannis78@cornell.edu,'79 Weeping Birch Point',0,1,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0,0,0,0,16000
36 Concordia,Kohen,ckohen@cocolog-nifty.com,'203 Northridge Circle',0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,1,0,0,0,0,39000
37 Nev,Yuryatin,nyuryatink@php.net,'65703 Ridgeview Drive',0,1,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0,0,0,0,45000
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